



SCOPE OF ACCREDITATION TO ISO 17025:2005
& ANSI/NCSL Z540-1-1994 & ANSI/NCSL Z540.3-2006

ACR TECHNICAL SERVICES, INC
 710 Industrial Park Drive
 Newport News, VA 23608
 Richard Hogan Phone: 757 890 0460

CALIBRATION

Valid To: March 31, 2017

Certificate Number: 2171.01

In recognition of the successful completion of the A2LA evaluation process, accreditation is granted to this laboratory to perform the following calibrations¹:

I. Acoustics & Vibration

Parameter/Range	Frequency	CMC ^{2,6} (±)	Comments
Vibration – Measuring Equipment (0.02 to 5000) pC/g (1 to 10 000) mV/g	5 Hz to 19 Hz	2.9 %	B&K 9610 w/ 4808
	(20 to 99) Hz	1.5 %	
	(0.1 to 1) kHz	1.5 %	
	(1 to 5) kHz	1.6 %	
	(5 to 10) kHz	2.6 %	
	100 Hz, 160 Hz	1.3 %	B&K 9610 w/ 4809

II. Chemical

Parameter/Equipment	Range	CMC ² (±)	Comments
pH Measuring Equipment	4.005 pH 6.989 pH 10.006 pH	0.02 pH 0.02 pH 0.02 pH	pH buffer solutions

Parameter/Equipment	Range	CMC ² (±)	Comments
Conductivity Liquid	1.01 µS 9.55 µS 99.8 µS 999 µS 9 988 µS 99 799 µS	0.43 µS 0.56 µS 4.3 µS 29 µS 1.1 % of value 1.3 % of value	Conductivity solutions
Gas Detection Equipment – CO ₂ O ₂ CO Propane (LEL)	5 % 10 % 20 % 30 % 5 % 15 % 19 % 21 % 0.01 % 25 %	1.1 % of value 1.3 % of value 1.2 % of value 1.2 % of value 1.1 % of value 1.2 % of value 1.2 % of value 1.2 % of value 1.1 % of value 2.5 % of value	Standard gas

III. Dimensional

Parameter/Equipment	Range	CMC ^{2, 5} (±)	Comments
Dial Indicators ^{3, 7}	Up to 1 in	93 µin	Universal calibrator
Micrometers and Calipers ^{3, 7}	(0.05 to 4) in (4 to 20) in	(13 + 8.3 <i>L</i>) µin (32 + 3.6 <i>L</i>) µin	Gage blocks
Anvil and Spindle Flatness on Micrometers and Calipers ³	2 in diameter Convex Concave	5.4 µin 5.6 µin	Optical flat and monochromatic light
Crimpers	(0.011 to 0.06) in (0.06 to 0.25) in	(270 – 630 <i>D</i>) µin (210 + 380 <i>D</i>) µin	Pin gages

IV. Electrical – DC/Low Frequency

Parameter/Equipment	Range	CMC ^{2, 4, 6} ($\pm$)	Comments
DC Voltage ³ – Generate	(0 to 220) mV 220 mV to 2.2 V (2.2 to 11) V (11 to 22) V (22 to 220) V (220 to 1100) V (1 to 10) kV	9.0 μ V/V + 0.4 μ V 5.2 μ V/V + 0.7 μ V 4.3 μ V/V + 2.5 μ V 4.3 μ V/V + 4 μ V 5.2 μ V/V + 40 μ V 7.3 μ V/V + 400 μ V 0.71 %	Fluke 5700A/EP with opt 3 Fluke 80K40 and Agilent 3458A, opt 002
DC Voltage – Fixed Points	0.1 V 1 V 10 V 100 V 1000 V 10 V Fixed	7.5 μ V/V 1.5 μ V/V 0.91 μ V/V 2.7 μ V/V 4.2 μ V/V 0.52 μ V/V	Fluke 732A & Fluke 752A Fluke 732A
DC Voltage ³ – Measure	(0 to 100) mV (0.1 to 1) V (1 to 10) V (10 to 100) V (0.1 to 1) kV* (1 to 10) kV	12 μ V/V + 0.3 μ V 4.5 μ V/V + 0.3 μ V 4.3 μ V/V + 0.5 μ V 4.7 μ V/V + 30 μ V 7.3 μ V/V + 100 μ V 0.71 %	Agilent 3458A, opt 002 ACAL (24hrs & $\pm$ 1°C); TCAL $\pm$ 5°C, & MATH NULL *add 12 (V _{in} /1000) ² ppm Fluke 80K40 and Agilent 3458A, opt 002
DC Current ³ – Measure Ohm's Law Method	100 nA to 1 μ A (1 to 10) μ A (10 to 100) μ A 100 μ A to 1 mA (1 to 10) mA (10 to 100) mA 100 mA to 1 A (10 to 100) nA 100 nA to 1 μ A (1 to 10) μ A (10 to 100) μ A 100 μ A to 1 mA (1 to 10) mA (10 to 100) mA 100 mA to 1 A (1 to 10) A (10 to 100) A	37 μ A/A + 40 pA 23 μ A/A + 40 pA 23 μ A/A + 100 pA 23 μ A/A + 800 pA 23 μ A/A + 5 nA 23 μ A/A + 50 nA 40 μ A/A + 500 nA 9.7 μ A/A 8.9 μ A/A 8.6 μ A/A 7.9 μ A/A 7.2 μ A/A 7.0 μ A/A 7.8 μ A/A 8.4 μ A/A 8.8 μ A/A 18 μ A/A	Agilent 3458A, opt 002 Fluke 5700A/EP Agilent 3458A, opt 002 ESI SR1060 Guildline 9211A

Parameter/Equipment	Range	CMC ^{2, 4, 6} (±)	Comments
DC Current ³ – Generate	(0 to 220) μ A	44 μ A/A + 6 nA	Fluke 5700A/EP
	(0.22 to 2.2) mA	39 μ A/A + 7 nA	
	(2.2 to 22) mA	37 μ A/A + 40 nA	
	(22 to 220) mA	50 μ A/A + 0.7 μ A	
	(0.22 to 2.2) A	93 μ A/A + 12 μ A	Fluke 5520A SC600
	(2.2 to 3) A	0.041 % + 6 μ A	
	(2.2 to 11) A	0.03 % + 1 mA	
	(11 to 20) A	0.034 % + 1 mA	Fluke 5700A/EP w/ Fluke 5220A SC600
	(0 to 100) A	0.046 % + 20 mA	Fluke 5700A/EP w/ Ballantine 1620A
Clamp-On Meters	(10 to 16.5) A (16.5 to 150) A (150 to 1025) A	0.3 % + 2 mA 0.27 % + 15 mA 0.29 % + 50 mA	Fluke 5700A/EP Fluke 5220A SC600 Fluke 5500ACOIL
Ohm's Law Method	(10 to 100) nA 100 nA to 1 μ A (1 to 10) μ A (10 to 100) μ A 100 μ A to 1 mA (1 to 10) mA (10 to 100) mA 100 mA to 1 A (1 to 10) A (10 to 100) A	9.7 μ A/A 8.9 μ A/A 8.6 μ A/A 7.9 μ A/A 7.2 μ A/A 7.0 μ A/A 7.8 μ A/A 8.4 μ A/A 8.8 μ A/A 18 μ A/A	Fluke 5700A/EP Agilent 3458A, opt 002 ESI SR1060 Guildline 9211A
Resistance ³ – Generate			
Fixed Values			
1 Ω steps	Up to 10 Ω	5.7 $\mu\Omega/\Omega$	ESI 242D ESI SR-104U ESI SR-1060
10 Ω steps	Up to 100 Ω	5.1 $\mu\Omega/\Omega$	
100 Ω steps	Up to 1000 Ω	4.0 $\mu\Omega/\Omega$	
1 k Ω steps	Up to 10 k Ω	2.0 $\mu\Omega/\Omega$	
10 k Ω steps	Up to 100 k Ω	2.5 $\mu\Omega/\Omega$	ESI SR-1050
100 k Ω steps	Up to 1 M Ω	4.3 $\mu\Omega/\Omega$	
1 M Ω steps	Up to 10 M Ω	5.5 $\mu\Omega/\Omega$	
10 M Ω steps	Up to 100 M Ω	6.0 $\mu\Omega/\Omega$	IET HRRS-B-4-1 M
100 M Ω steps	Up to 1 G Ω	0.33 % - 0.21M Ω	
1 G Ω steps	Up to 10 G Ω	0.62 %	
Fixed Values			
10 k Ω Fixed	10 k Ω	1.2 $\mu\Omega/\Omega$	ESI SR-104U
1 Ω Fixed	1 Ω	1.6 $\mu\Omega/\Omega$	Leeds & Northrop 4020B

Parameter/Equipment	Range	CMC ^{2, 4, 6} (±)	Comments
Resistance ³ – Measure	Up to 10 Ω Up to 100 Ω Up to 1000 Ω Up to 10 kΩ Up to 100 kΩ Up to 1 MΩ Up to 10 MΩ Up to 100 MΩ Up to 1 GΩ	19 μΩ/Ω + 50 μΩ 15 μΩ/Ω + 500 μΩ 13 μΩ/Ω + 500 μΩ 12 μΩ/Ω + 5 mΩ 12 μΩ/Ω + 50 mΩ 18 μΩ/Ω + 2 Ω 58 μΩ/Ω + 100 Ω 0.057 % + 1 kΩ 0.57 % + 10 kΩ	Agilent 3458A, opt 002

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
AC Voltage, Fixed Points			
2 mV	10 Hz 20 Hz 100 Hz 1 kHz 10 kHz 20 kHz 50 kHz 100 kHz 300 kHz 500 kHz 1 MHz	0.04 % 0.033 % 0.032 % 0.031 % 0.031 % 0.031 % 0.032 % 0.043 % 0.054 % 0.063 % 0.13 %	Fluke 792A
6 mV	10 Hz 20 Hz 100 Hz 1 kHz 10 kHz 20 kHz 50 kHz 100 kHz 300 kHz 500 kHz 1 MHz	0.023 % 0.022 % 0.016 % 0.016 % 0.016 % 0.016 % 0.021 % 0.028 % 0.04 % 0.046 % 0.065 %	
20 mV	10 Hz 20 Hz 100 Hz 1 kHz 10 kHz 20 kHz 50 kHz 100 kHz 300 kHz 500 kHz 1 MHz	0.011 % 69 μV/V 66 μV/V 65 μV/V 65 μV/V 65 μV/V 93 μV/V 0.016 % 0.026 % 0.036 % 0.042 %	

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
AC Voltage, Fixed Points (cont)			
60 mV	10 Hz	0.011 %	Fluke 792A
	20 Hz	70 µV/V	
	100 Hz	50 µV/V	
	1 kHz	50 µV/V	
	10 kHz	50 µV/V	
	20 kHz	57 µV/V	
	50 kHz	77 µV/V	
	100 kHz	0.016 %	
	300 kHz	0.028 %	
	500 kHz	0.037 %	
	1 MHz	0.044 %	
200 mV	10 Hz	39 µV/V	
	20 Hz	27 µV/V	
	100 Hz	21 µV/V	
	1 kHz	21 µV/V	
	10 kHz	21 µV/V	
	20 kHz	22 µV/V	
	50 kHz	26 µV/V	
	100 kHz	44 µV/V	
	300 kHz	82 µV/V	
	500 kHz	0.012 %	
	1 MHz	0.019 %	
600 mV	10 Hz	30 µV/V	
	20 Hz	24 µV/V	
	100 Hz	14 µV/V	
	1 kHz	13 µV/V	
	10 kHz	13 µV/V	
	20 kHz	14 µV/V	
	50 kHz	14 µV/V	
	100 kHz	18 µV/V	
	300 kHz	36 µV/V	
	500 kHz	39 µV/V	
	1 MHz	71 µV/V	
2 V	10 Hz	30 µV/V	
	20 Hz	24 µV/V	
	100 Hz	12 µV/V	
	1 kHz	11 µV/V	
	10 kHz	11 µV/V	
	20 kHz	11 µV/V	
	50 kHz	11 µV/V	
	100 kHz	17 µV/V	
	300 kHz	28 µV/V	
	500 kHz	33 µV/V	
	1 MHz	50 µV/V	

Parameter/Range	Frequency	CMC ^{2,4} (±)	Comments
AC Voltage, Fixed Points (cont)			
6 V	10 Hz	30 µV/V	Fluke 792A
	20 Hz	23 µV/V	
	100 Hz	12 µV/V	
	1 kHz	11 µV/V	
	10 kHz	11 µV/V	
	20 kHz	11 µV/V	
	50 kHz	12 µV/V	
	100 kHz	16 µV/V	
	300 kHz	28 µV/V	
	500 kHz	33 µV/V	
	1 MHz	50 µV/V	
20 V	10 Hz	30 µV/V	
	20 Hz	23 µV/V	
	100 Hz	13 µV/V	
	1 kHz	13 µV/V	
	10 kHz	13 µV/V	
	20 kHz	13 µV/V	
	50 kHz	13 µV/V	
	100 kHz	19 µV/V	
	300 kHz	29 µV/V	
	500 kHz	35 µV/V	
	1 MHz	50 µV/V	
60 V	10 Hz	28 µV/V	
	20 Hz	23 µV/V	
	100 Hz	14 µV/V	
	1 kHz	13 µV/V	
	10 kHz	13 µV/V	
	20 kHz	13 µV/V	
	50 kHz	15 µV/V	
	100 kHz	20 µV/V	
	300 kHz	36 µV/V	
200 V	10 Hz	41 µV/V	
	20 Hz	25 µV/V	
	100 Hz	15 µV/V	
	1 kHz	15 µV/V	
	10 kHz	15 µV/V	
	20 kHz	15 µV/V	
	50 kHz	21 µV/V	
	100 kHz	26 µV/V	
600 V	100 Hz	22 µV/V	
	1 kHz	17 µV/V	
	10 kHz	19 µV/V	
	20 kHz	23 µV/V	
	50 kHz	45 µV/V	
	100 kHz	73 µV/V	
1000 V	100 Hz	23 µV/V	
	1 kHz	22 µV/V	
	10 kHz	23 µV/V	
	20 kHz	24 µV/V	
	50 kHz	51 µV/V	
	100 kHz	76 µV/V	

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
AC Voltage Flatness – Generate			
1 mV	(10 to 30) Hz (30 to 120) Hz (0.12 to 1.2) kHz (1.2 to 120) kHz (120 to 500) kHz (0.12 to 1.2) MHz (1.2 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.062 % 0.038 % 0.038 % 0.038 % 0.046 % 0.05 % 0.05 % 0.071 % 0.23 % 0.74 %	Fluke 5700A/EP/03
3 mV	(10 to 30) Hz (30 to 120) Hz (0.12 to 1.2) kHz (1.2 to 120) kHz (120 to 500) kHz (0.12 to 1.2) MHz (1.2 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.038 % 0.024 % 0.024 % 0.024 % 0.033 % 0.033 % 0.036 % 0.061 % 0.11 % 0.19 %	
10 mV	(10 to 30) Hz (30 to 120) Hz (0.12 to 1.2) kHz (1.2 to 120) kHz (120 to 500) kHz (0.12 to 1.2) MHz (1.2 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.033 % 0.021 % 0.021 % 0.021 % 0.028 % 0.029 % 0.033 % 0.058 % 0.1 % 0.18 %	
30 mV	(10 to 30) Hz (30 to 120) Hz (0.12 to 1.2) kHz (1.2 to 120) kHz (120 to 500) kHz (0.12 to 1.2) MHz (1.2 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.035 % 0.02 % 0.02 % 0.02 % 0.025 % 0.029 % 0.032 % 0.057 % 0.1 % 0.18 %	

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
AC Voltage Flatness – Generate			
100 mV	(10 to 30) Hz (30 to 120) Hz (0.12 to 1.2) kHz (1.2 to 120) kHz (120 to 500) kHz (0.12 to 1.2) MHz (1.2 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.035 % 0.02 % 0.02 % 0.02 % 0.025 % 0.029 % 0.032 % 0.057 % 0.1 % 0.18 %	Fluke 5700A/EP/03
300 mV	(10 to 30) Hz (30 to 120) Hz (0.12 to 1.2) kHz (1.2 to 120) kHz (120 to 500) kHz (0.12 to 1.2) MHz (1.2 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.026 % 0.018 % 0.018 % 0.018 % 0.027 % 0.028 % 0.033 % 0.053 % 0.1 % 0.17 %	
1 V	(10 to 30) Hz (30 to 120) Hz (0.12 to 1.2) kHz (1.2 to 120) kHz (120 to 500) kHz (0.12 to 1.2) MHz (1.2 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.031 % 0.017 % 0.016 % 0.015 % 0.022 % 0.026 % 0.03 % 0.055 % 0.1 % 0.17 %	
3 V	(10 to 30) Hz (30 to 120) Hz (0.12 to 1.2) kHz (1.2 to 120) kHz (120 to 500) kHz (0.12 to 1.2) MHz (1.2 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.036 % 0.015 % 0.015 % 0.015 % 0.021 % 0.028 % 0.031 % 0.055 % 0.1 % 0.17 %	

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
AC Voltage ³ – Generate			
(0.22 to 2.2) mV	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.024 % + 4 µV 90 µV/V + 4 µV 80 µV/V + 4 µV 0.02 % + 4 µV 0.05 % + 4 µV 0.11 % + 10 µV 0.14 % + 10 µV 0.27 % + 20 µV	Fluke 5700A/EP
(2.2 to 22) mV	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.024 % + 4 µV 90 µV/V + 4 µV 80 µV/V + 4 µV 0.02 % + 4 µV 0.05 % + 4 µV 0.11 % + 10 µV 0.14 % + 10 µV 0.27 % + 20 µV	
(22 to 220) mV	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.024 % + 4 µV 90 µV/V + 4 µV 80 µV/V + 4 µV 0.02 % + 4 µV 0.05 % + 4 µV 0.11 % + 10 µV 0.14 % + 10 µV 0.27 % + 20 µV	
(0.22 to 2.2) V	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.024 % + 40 µV 90 µV/V + 15 µV 50 µV/V + 8 µV 80 µV/V + 10 µV 0.011 % + 30 µV 0.042 % + 80 µV 0.1 % + 200 µV 0.17 % + 300 µV	

Parameter/Range	Frequency	CMC ^{2,4} (±)	Comments
AC Voltage ³ – Generate (cont)			
(2.2 to 22) V	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.024 % + 0.4 mV 90 µV/V + 0.15 mV 50 µV/V + 0.05 mV 80 µV/V + 0.1 mV 0.01 % + 0.2 mV 0.028 % + 0.6 mV 0.1 % + 2 mV 0.15 % + 3.2 mV	Fluke 5700A/EP
(22 to 220) V	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.024 % + 0.4 mV 90 µV/V + 0.15 mV 50 µV/V + 0.05 mV 80 µV/V + 0.1 mV 0.015 % + 0.2 mV 0.09 % + 0.6 mV 0.44 % + 2 mV 0.8 % + 3.2 mV	
(220 to 1100) V	(15 to 50) Hz 50 Hz to 1 kHz	0.03 % + 16 mV 0.007 % + 3.5 mV	
AC Voltage ³ – Measure			
Up to 2.2 mV	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.17 % + 1.3 µV 0.074 % + 1.3 µV 0.042 % + 1.3 µV 0.081 % + 2 µV 0.12 % + 2.5 µV 0.23 % + 4 µV 0.24 % + 8 µV 0.35 % + 8 µV	Fluke 5790A opt.003 uncertainty of wideband is for flatness relative to 1 kHz
Wideband	(1 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.07 % + 1 µV 0.17 % + 1 µV 0.3 % + 1 µV 0.7 % + 2 µV	
(2.2 to 7) mV	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.085 % + 1.3 µV 0.037 % + 1.3 µV 0.021 % + 1.3 µV 0.04 % + 2 µV 0.06 % + 2.5 µV 0.12 % + 4 µV 0.13 % + 8 µV 0.23 % + 8 µV	

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
AC Voltage ³ – Measure (cont)			
(2.2 to 7) mV Wideband	(1 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.07 % + 1 µV 0.1 % + 1 µV 0.17 % + 1 µV 0.37 % + 1 µV	Fluke 5790A opt.003 uncertainty of wideband is for flatness relative to 1 kHz
(7 to 22) mV	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.029 % + 1.3 µV 0.019 % + 1.3 µV 0.011 % + 1.3 µV 0.021 % + 2 µV 0.031 % + 2.5 µV 0.081 % + 4 µV 0.089 % + 8 µV 0.17 % + 8 µV	
Wideband	(1 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.07 % 0.1 % 0.17 % 0.37 %	
(22 to 70) mV	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.024 % + 1.5 µV 0.012 % + 1.5 µV 0.007 % + 1.5 µV 0.013 % + 2 µV 0.026 % + 2.5 µV 0.051 % + 4 µV 0.067 % + 8 µV 0.11 % + 8 µV	
Wideband	(1 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.05 % 0.1 % 0.15 % 0.35 %	
(70 to 220) mV	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.021 % + 1.5 µV 0.009 % + 1.5 µV 0.004 % + 1.5 µV 0.007 % + 2 µV 0.016 % + 2.5 µV 0.025 % + 4 µV 0.038 % + 8 µV 0.1 % + 8 µV	
Wideband	(1 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.05 % 0.1 % 0.15 % 0.35 %	

Parameter/Range	Frequency	CMC ^{2, 4, 6} ($\pm$)	Comments
AC Voltage ³ – Measure (cont)			
(220 to 700) mV	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.02 % + 1.5 μ V 70 μ V/V + 1.5 μ V 20 μ V/V + 1.5 μ V 50 μ V/V + 2 μ V 70 μ V/V + 2.5 μ V 0.016 % + 4 μ V 0.026 % + 8 μ V 0.09 % + 8 μ V	Fluke 5790A opt.003 uncertainty of wideband is for flatness relative to 1 kHz
Wideband	(1 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.05 % 0.1 % 0.15 % 0.35 %	
700 mV to 2.2 V	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.021 % 80 μ V/V 30 μ V/V 50 μ V/V 80 μ V/V 0.018 % 0.03 % 0.1 %	
Wideband	(1 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.05 % 0.1 % 0.15 % 0.35 %	
(2.2 to 7) V	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz 500 kHz to 1 MHz	0.02 % 70 μ V/V 20 μ V/V 50 μ V/V 70 μ V/V 0.016 % 0.026 % 0.09 %	
Wideband	(1 to 2) MHz (2 to 10) MHz (10 to 20) MHz (20 to 30) MHz	0.05 % 0.1 % 0.15 % 0.35 %	
(7 to 22) V	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz (0.5 to 1) MHz	0.02 % 70 μ V/V 20 μ V/V 50 μ V/V 80 μ V/V 0.019 % 0.04 % 0.12 %	

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
AC Voltage ³ – Measure (cont)			
(22 to 70) V	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz (0.5 to 1) MHz	0.02 % 70 µV/V 30 µV/V 50 µV/V 90 µV/V 0.02 % 0.041 % 0.12 %	Fluke 5790A opt.003 uncertainty of wideband is for flatness relative to 1 kHz
(70 to 220) V	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (300 to 500) kHz	0.02 % 70 µV/V 30 µV/V 70 µV/V 0.01 % 0.021 % 0.05 %	
(220 to 700) V	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz	0.02 % 0.01 % 40 µV/V 0.013 % 0.05 %	
(700 to 1000) V	(10 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz	0.02 % 0.01 % 0.004 % 0.013 % 0.05 %	
(0 to 10) mV	(1 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (0.3 to 1) MHz	0.03 % + 3 µV 0.02 % + 1.1 µV 0.03 % + 1.1 µV 0.1 % + 1.1 µV 0.5 % + 1.1 µV 4 % + 2 µV 1.2 % + 5 µV	
(10 to 100) mV	(1 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (0.3 to 1) MHz	70 µV/V + 4 µV 70 µV/V + 2 µV 0.014 % + 2 µV 0.03 % + 2 µV 0.08 % + 2 µV 0.3 % + 10 µV 1 % + 10 µV	

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
AC Voltage ³ – Measure (cont)			
(0.1 to 10) V	(1 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (0.3 to 1) MHz	70 µV/V + 0.4 mV 70 µV/V + 0.2 mV 0.014 % + 0.2 mV 0.03 % + 0.2 mV 0.08 % + 0.2 mV 0.3 % + 1 mV 1 % + 1 mV	Agilent 3458A, opt 002 *synchronous sub-sampled mode
(10 to 100) V	(1 to 40) Hz 40 Hz to 20 kHz (20 to 50) kHz (50 to 100) kHz (100 to 300) kHz (0.3 to 1) MHz	0.02 % + 4 mV 0.02 % + 2 mV 0.035 % + 2 mV 0.12 % + 2 mV 0.4 % + 10 mV 1.5 % + 10 mV	
(100 to 1000) V	(1 to 40) Hz 40 Hz to 1 kHz (1 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.04 % + 40 mV 0.04 % + 20 mV 0.06 % + 20 mV 0.12 % + 20 mV 0.3 % + 20 mV	
(0 to 40) kV _{p-p} (0 to 28) kV _{rms}	60 Hz 60 Hz	5 % 5 %	Fluke 80K40
AC Voltage Ratio ³ – Measure			
(0.0 to 0.1) V	(50 to 400) Hz (0.4 to 1) kHz (1 to 3) kHz (3 to 5) kHz (5 to 10) kHz	7.4 µV/V 7.8 µV/V 13 µV/V 13 µV/V 29 µV/V	DT72A w/ Fluke 5520A

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
AC Current ³ – Generate			
(0 to 220) µA	(10 to 20) Hz (20 to 40) Hz 40 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.025 % + 16 nA 0.016 % + 10 nA 0.012 % + 8 nA 0.028 % + 12 nA 0.11 % + 65 nA	Fluke 5700A/EP
(0.22 to 2.2) mA	(10 to 20) Hz (20 to 40) Hz 40 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.025 % + 40 nA 0.016 % + 35 nA 0.012 % + 35 nA 0.02 % + 110 nA 0.11 % + 650 nA	
(2.2 to 22) mA	(10 to 20) Hz (20 to 40) Hz 40 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.025 % + 400 nA 0.016 % + 350 nA 0.012 % + 350 nA 0.02 % + 550 nA 0.11 % + 5 µA	
(22 to 220) mA	(10 to 20) Hz (20 to 40) Hz 40 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.025 % + 4 µA 0.016 % + 3.5 µA 0.012 % + 2.5 µA 0.02 % + 3.5 µA 0.11 % + 10 µA	
(0.22 to 2.2) A	10 Hz to 1 kHz (1 to 5) kHz (5 to 10) kHz	0.026 % + 35 µA 0.045 % + 80 µA 0.7 % + 160 µA	
(29 to 330) µA (0.33 to 3.3) mA (3.3 to 33) mA (33 to 330) mA (1.1 to 3) A	(10 to 30) kHz (10 to 30) kHz (10 to 30) kHz (10 to 30) kHz (5 to 10) kHz	1.6 % + 0.4 µA 1 % + 0.6 µA 0.4 % + 4 µA 0.4 % + 200 µA 2.5 % + 5 mA	Fluke 5520A SC600
(2.2 to 22) A	(10 to 20) Hz (20 to 40) Hz 40 Hz to 1 kHz (1 to 5) kHz*	0.055 % + 1 mA 0.051 % + 1 mA 0.05 % + 1 mA 0.05 % + 1 mA	Fluke 5700A/EP Fluke 5520A SC600 *multiply by f in kHz

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
AC Current ³ – Generate (cont)			
Clamp-On Meters			
(10 to 16.5) A	(45 to 65) Hz (65 to 440) Hz	0.33 % + 0.025 A 0.82 % + 0.027 A	Fluke 5700A/EP Fluke 5220A SC600 Fluke 5500A coil
(16.5 to 150) A	(45 to 65) Hz (65 to 440) Hz	0.3 % + 0.025 A 0.78 % + 0.027 A	
(150 to 1025) A	(45 to 65) Hz (65 to 440) Hz	0.4 % + 0.025 A 0.87 % + 0.027 A	
AC Current ³ – Measure			
(0 to 100) µA	(10 to 20) Hz (20 to 45) Hz 45 Hz to 5 kHz	0.4 % + 30 nA 0.15 % + 30 nA 0.06 % + 30 nA	Agilent 3458A opt002
(0.1 to 1) mA	(10 to 20) Hz (20 to 45) Hz (45 to 100) Hz (0.1 to 5) kHz	0.4 % + 0.2 µA 0.15 % + 0.2 µA 0.06 % + 0.2 µA 0.03 % + 0.2 µA	
(1 to 100) mA	(10 to 20) Hz (20 to 45) Hz (45 to 100) Hz (0.1 to 5) kHz (5 to 20) kHz (20 to 50) kHz (50 to 100) kHz	0.4 % + 20 µA 0.15 % + 20 µA 0.06 % + 20 µA 0.03 % + 20 µA 0.06 % + 20 µA 0.4 % + 40 µA 0.6 % + 150 µA	
(0.1 to 1) A	(10 to 20) Hz (20 to 45) Hz (45 to 100) Hz (0.1 to 5) kHz (5 to 20) kHz (20 to 50) kHz	0.4 % + 0.2 mA 0.16 % + 0.2 mA 0.08 % + 0.2 mA 0.1 % + 0.2 mA 0.3 % + 0.2 mA 1 % + 0.4 mA	
(0 to 550) µA	(5 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz	0.02 % 0.008 % 0.005 %	Fluke 5790A w/ A40
550 µA to 5.5 mA	(5 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz	0.02 % 0.008 % 0.005 %	

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
AC Current ³ – Measure (cont)			
(5.5 to 22) mA	(5 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz	0.02 % 0.008 % 0.005 %	Fluke 5790A w/ A40
(22 to 550) mA	(5 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz	0.02 % 0.008 % 0.005 %	
550 mA to 2.2 A	(5 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz	0.02 % 0.009 % 0.008 %	
(2.2 to 11) A	(5 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz	0.022 % 0.013 % 0.021 %	
(11 to 20) A	(5 to 20) Hz (20 to 40) Hz 40 Hz to 20 kHz	0.038 % 0.034 % 0.034 %	
Inductance ³ – Measure			
0.01 µH to 4.1 H	(12 to 100) Hz 100 Hz to 1 kHz (1 to 10) kHz (10 to 100) kHz	0.049 % 0.029 % 0.059 % 0.23 %	GenRad 1693 Digibridge *Slow measure rate, Averaging, Open/Short
(4.1 to 99 999) H	(12 to 100) Hz 100 Hz to 1 kHz (1 to 20) kHz	0.049 % 0.029 % 0.21 %	
Inductance ³ – Generate			
100 mH fixed	100 Hz 1 kHz	0.023 % 0.023 %	General ratio 1482-L
Capacitance ³ – Measure			
0.00001 pF to 6.4 nF	(12 to 100) Hz 100 Hz to 1 kHz (1 to 20) kHz	0.049 % 0.029 % 0.21 %	GenRad 1693 Digibridge *Slow measure rate, Averaging, Open/Short

Parameter/Range	Frequency	CMC ^{2, 4, 6} (±)	Comments
Phase ³ – Measure (cont) (0 to 360)°			
1:1 Resistive	1 Hz to 1 kHz (1 to 50) kHz	0.002° 0.003°	Clarke-Hess 5002 Agilent 3458A opt. 002
1:1 Capacitive	(1 to 50) kHz (50 to 200) kHz	0.003° 0.009°	
10:1 Capacitive	(1 to 50) kHz (50 to 200) kHz	0.003° 0.009°	
100:1 Capacitive	(1 to 50) kHz (50 to 200) kHz	0.003° 0.009°	
Distortion ³ – Measure			
0.3 to 100 % < 30 V	10 Hz to 1 MHz (1 to 3) MHz	3 % of full scale 7.1 % of full scale	HP 334A
(30 to 300) V	10 Hz to 300 kHz (300 to 500) kHz (0.5 to 3) MHz	3 % of full scale 7.1 % of full scale 13 % of full scale	
0.1 % < 30 V	(10 to 20) Hz (20 to 30) Hz 30 Hz to 300 kHz (300 to 500) kHz (0.5 to 1.2) MHz	13 % of full scale 7.1 % of full scale 3 % of full scale 7.1 % of full scale 13 % of full scale	Agilent 8903B
0.001 % to 100 % 50 mV to 300 V	20 Hz to 20 kHz (20 to 100) kHz	1.2 dB 2.3 dB	
Sine Wave Flatness ³ – Measure	1 MHz 10 MHz 20 MHz 50 MHz 100 MHz	0.087 % 0.14 % 0.19 % 0.58 % 1.3 %	Precision Measurements EL 1200

Parameter/Range	Frequency	CMC ^{2, 4, 6} ($\pm$)	Comments
Oscilloscopes ³ –			
Square Wave Signal			
10 Hz to 10 kHz			
50 Ω Impedance	1 mV to 6.6 V _{pk-pk}	0.25 % + 40 μ V	Fluke 5520A SC600
1 M Ω Impedance	1 mV to 130 V _{pk-pk}	0.1 % + 40 μ V	
Level Sine Wave	5 mV to 5.5 V	2 % + 300 μ V	
Square Wave Signal	50 kHz to 100 MHz	3.5 % + 300 μ V	
	(100 to 300) MHz	4 % + 300 μ V	
	(300 to 600) MHz	6 % + 300 μ V	
Amplitude (Reference, 50 kHz)	50 kHz to 100 MHz	1.5 % + 100 μ V	
	(100 to 300) MHz	2 % + 100 μ V	
	(300 to 600) MHz	4 % + 100 μ V	
Flatness (Reference, 50 kHz)	5 s to 50 ms	25 μ s/s	
	20 ms to 100 ns	2.5 μ s/s	
Time Markers (5-2-1 Sequence) into a 50 W Load	(5 to 2) ns	2.5 μ s/s	Fluke 5520A SC600
	10 ns	2.5 μ s/s	
	(50 to 20) ns	2.5 μ s/s	
Rise Time 2.5 mV to 2.5 V	(24 to 300) ps	27 %	

Parameter/Equipment	Range	CMC ² (±)	Comments
Electrical Calibration of Thermocouple Indicating Systems ³ –			
Type B	(0 to 800) °C (800 to 1820) °C	0.16 °C 0.17 °C	Fluke 5700A/EP and Omega TRC III ice point reference
Type C	(0 to 2320) °C	0.2 °C	
Type N	(-270 to 0) °C (0 to 1300) °C	0.18 °C 0.28 °C	
Type E	(-270 to 0) °C (0 to 1300) °C	0.17 °C 0.2 °C	
Type R	(-50 to 0) °C (0 to 1767) °C	0.17 °C 0.16 °C	
Type S	(-50 to 0) °C (0 to 1767) °C	0.17 °C 0.16 °C	
Type J	(-210 to 0) °C (0 to 1200) °C	0.28 °C 0.18 °C	
Type K	(-210 to 0) °C (0 to 1200) °C	0.28 °C 0.18 °C	
Type T	(-270 to 0) °C (0 to 400) °C	0.23 °C 0.18 °C	
Type U	(-200 to 0) °C (0 to 600) °C	0.28 °C 0.2 °C	
Electrical Calibration of RTDs ³ –			
Pt 385, 1000 Ω	(-200 to 0) °C (0 to 100) °C (100 to 260) °C (260 to 300) °C (300 to 600) °C (600 to 630) °C	0.04 °C 0.05 °C 0.06 °C 0.07 °C 0.08 °C 0.24 °C	Fluke 5520A SC600
PtNi 385, 120 Ω	(-80 to 100) °C (100 to 260) °C	0.08 °C 0.15 °C	
Cu 427, 10 Ω	(-100 to 260) °C	0.34 °C	

V. Electrical – RF/Microwave

Parameter/Range	Frequency	CMC ^{2, 6} ($\pm$)	Comments
RF Power – Generate			
Sine Wave (50 Ω) 1 mV to 3 V _{p-p} (-56.02 to 13.52) dB	0.001 Hz to 100 kHz 100 kHz to 20 MHz	0.11 dB of output 0.46 dB of output	HP 3325A opt 001, 002
(3 to 10) V _{p-p} (13.52 to 23.98) dB	0.001 Hz to 100 kHz 100 kHz to 20 MHz	0.23 dB of output 0.58 dB of output	
(25 to 10) dB (10 to -10) dB (-10 to -60) dB (-60 to -110) dB	10 MHz to 2 GHz	1.2 dB of output 0.6 dB of output 0.9 dB of output 1.4 dB of output	HP 83630A opt 001, 008, H53
(25 to 10) dB (10 to -10) dB (-10 to -60) dB (-60 to -110) dB	(2 to 18) GHz	1.3 dB of output 0.7 dB of output 1.0 dB of output 1.5 dB of output	
1 mW	50 MHz	1.0 %	HP 436A
RF Power – Measure			
1 μ W to 100 mW (-30 to +20) dB	(100 to 300) kHz 300 kHz to 1 MHz 1 MHz to 2 GHz (2 to 4.2) GHz	0.05 dB 0.049 dB 0.051 dB 0.05 dB	HP 436A w/ 8482A
1 μ W to 100 mW (-30 to +20) dB	10 MHz to 3 GHz (3 to 10) GHz (10 to 15) GHz (15 to 18) GHz	0.045 dB 0.049 dB 0.054 dB 0.058 dB	HP 436A w/ 8481A
1 mW	50 MHz	0.69 %	HP 432A, 478AH73, and 34401A

Parameter/Range	Frequency	CMC ^{2, 6} (±)	Comments
Tuned RF Power – Measure (0 to -20) dB (-10 to -20) dB (-20 to -30) dB (-30 to -40) dB (-40 to -50) dB (-50 to -60) dB (-60 to -70) dB (-70 to -80) dB (-80 to -90) dB (-90 to -100) dB (-100 to -110) dB (-110 to -120) dB	100 kHz to 18 GHz	0.03 dB 0.06 dB 0.08 dB 0.11 dB 0.14 dB 0.16 dB 0.20 dB 0.22 dB 0.24 dB 0.26 dB 0.32 dB 0.37 dB	HP 8902A, HP 11722A, HP 11792A
Attenuation – Generate SMA, 1 dB step (1 to 11) dB SMA, 10 dB step (10 to 110) dB	(DC to 18 GHz) (DC to 18 GHz)	1.2 % + 0.2 dB 0.42 % + 0.27 dB	Agilent 8494H Agilent 8496H
Frequency Modulation – Generate Rate: 20 Hz to 200 kHz Dev.: ≤ 400 kHz peak Rate: 100 kHz to 8 MHz Dev: < 500 kHz peak	(11 to 13.5) MHz (88 to 108) MHz (352 to 432) MHz (2 to 18) GHz	0.3 % of output 10 % of output	Agilent 11715A Agilent 83630A
Frequency Modulation – Measure Rate: 20 Hz to 10 kHz Dev: ≤ 40 kHz Rate: 50 Hz to 100 kHz Dev: ≤ 400 kHz Rate: 20 Hz to 200 kHz Dev: ≤ 400 kHz	(0.25 to 10) MHz 10 MHz to 18 GHz 10 MHz to 18 GHz	2 % + 1 digit 1 % + 1 digit 5 % + 1 digit	HP 8902A w/ 11793A

Parameter/Range	Frequency	CMC ^{2, 6} (±)	Comments
Amplitude Modulation – Measure			
Rate: 150 kHz to 10 MHz Depths: (5 to 99) %	50 Hz to 10 kHz	2 % + 1 digit	HP 8902A w/ 11793A
Rate: 150 kHz to 10 MHz Depths: To 99 %	20 Hz to 10 kHz	3 % + 1 digit	
Rate: (10 to 1300) MHz Depths: (5 to 99) %	50 Hz to 50 kHz	1 % + 1 digit	
Rate: (10 to 1300) MHz Depths: To 99 %	20 Hz to 100 kHz	3 % + 1 digit	
Rate: (1.3 to 18) GHz Depths: (5 to 99) %	50 Hz to 50 kHz	1 % + 1 digit	
Rate: (1.3 to 18) GHz Depths: To 99 %	20 Hz to 100 kHz	3 % + 1 digit	
Amplitude Modulation – Generate			
Rate: 50 Hz to 100 kHz Depths: 5 % to 99 %	(11 to 13.5) MHz	0.12 % of output	Agilent 11715A
Rate: DC to 100 kHz Depths: 5 % to 99 %	10 MHz to 18 GHz	5 % of output	Agilent 83630A
Phase Modulation – Measure			
150 kHz to <10 MHz Carrier	200 Hz to 10 kHz	4.0 % + 1 digit	HP 8902A w/ 11793A
10 MHz to 18.0 GHz Carrier	200 Hz to 20 kHz	3.2 % + 1 digit	

Parameter/Equipment	Range	CMC ^{2, 7} ($\pm$)	Comments
Reflection S_{11}/S_{22} – Measure 50 MHz to 18 GHz (3.5 mm)	(0.56 to 1) lin (0.32 to 0.56) lin (0.1 to 0.32) lin (0.032 to 0.1) lin (0 to 0.032) lin	(0.005 to 0.063) lin (0.8 to 1.7) deg (0.004 to 0.035) lin (1.7 to 2.1) deg (0.004 to 0.024) lin (2.1 to 5.1) deg (0.004 to 0.008) lin (5.1 to 8.2) deg (0.004 to 0.008) lin (8.2 to 180) deg	HP 8720C w/ 85052D
Transmission S_{12}/S_{21} – Measure 50 MHz to 18 GHz (3.5 mm)	(0 to 12) dB (12 to 20) dB (20 to 30) dB (30 to 40) dB (40 to 50) dB (50 to 60) dB	(0.056 to 0.17) dB (0.52 to 1.8) deg (0.079 to 0.18) dB (0.52 to 1.8) deg (0.081 to 0.26) dB (0.53 to 1.8) deg (0.089 to 0.26) dB (0.59 to 1.8) deg (0.092 to 0.27) dB (0.61 to 2.6) deg (0.095 to 1.1) dB (0.63 to 7.8) deg	HP 8720C w/ 85052D
Transmission S_{12}/S_{21} – Measure 50 MHz to 18 GHz (3.5 mm)	(60 to 70) dB (70 to 80) dB	(0.13 to 3.0) dB (0.79 to 25) deg (0.25 to 7.3) dB (1.7 to 180) deg	HP 8720C w/ 85052D

VI. Mechanical

Parameter/Equipment	Range	CMC ^{2, 6, 7} ($\pm$)	Comments
Pressure and Vacuum Gauges ³ – Pneumatic	(0.001 to 24) in·H ₂ O	0.014 % + 0.0018 in·H ₂ O	Dwyer 1425-24
Vacuum	(30 to 100) microns (100 to 1000) microns	4.3 % + 1.0 microns 6.4 % – 1.1 microns	Leybold CTR 90
Pressure and Vacuum Gauges ³ – Pneumatic	(0.2 to 25) psia (0.2 to 25) psig (25 to 1000) psig (1000 to 12 140) psig	16 parts in 10 ⁶ psia 16 parts in 10 ⁶ psig 14 parts in 10 ⁶ psig 45 parts in 10 ⁶ psig	Ruska 2465-725 Ruska 2465-729 Ruska 2400-735-00 Ruska 2402
Hydraulic	(6 to 2428) psig (30 to 12 140) psig	28 parts in 10 ⁶ psig 39 parts in 10 ⁶ psig	Ruska 2400-736-00 Ruska 2400-735-00
Air/Nitrogen Flow ³ – (0.01 to 1000) SLPM (0.0004 to 35) SCFM	(1 to 10) sccm (10 to 30) sccm (30 to 100) sccm (100 to 300) sccm (0.3 to 1.0) SLPM (1.0 to 3.0) SLPM (3.0 to 10) SLPM (10 to 30) SLPM (30 to 100) SLPM (100 to 300) SLPM (300 to 1000) SLPM	0.57 % + 0.018 sccm 0.3 % + 0.09 sccm 0.34 % + 0.29 sccm 0.4 % + 1.0 sccm 0.43 % + 0.003 slpm 0.4 % + 0.012 slpm 0.46 % + 0.024 slpm 0.5 % + 0.06 slpm 0.54 % + 0.26 slpm 0.65 % + 1.3 slpm 0.8 % + 2.0 slpm	CME FCS-8A and Laminar flow elements
Torque ³ – Measure	(5 to 15) in·ozf (15 to 200) in·ozf (10 to 20) in·lbf (20 to 100) in·lbf (10 to 125) ft·lbf (50 to 600) ft·lbf (100 to 1000) ft·lbf	0.16 % 0.16 % 0.2 % + 0.1 in·lbf 0.11 % + 0.12 in·lbf 0.16 % 0.16 % 0.16 %	CDI 2000-4-02 CDI 2000-5-02 Mountz TL-100i CDI 2000-10-02 CDI 2000-12-02 CDI 2000-13-02

Parameter/Equipment	Range	CMC ^{2, 6} (±)	Comments
Piston Operated Volumetric Apparatus (POVA) ^{3, 7} – 10 µL to 200 mL	10 µL 20 µL 25 µL 30 µL 50 µL 100 µL 120 µL 150 µL 200 µL 250 µL 300 µL 500 µL 600 µL 1000 µL 1.2 mL 1.25 mL 2 mL 2.5 mL 5 mL 10 mL 12.5 mL 25 mL 50 mL 100 mL 200 mL	1.6 µL 1.6 µL 1.6 µL 1.6 µL 1.6 µL 1.6 µL 1.6 µL 1.6 µL 1.6 µL 1.6 µL 1.6 µL 1.6 µL 1.6 µL 1.6 µL 0.0019 mL 0.002 mL 0.0031 mL 0.0039 mL 0.0076 mL 0.015 mL 0.016 mL 0.033 mL 0.065 mL 0.13 mL 0.26 mL	Ohaus AR2140 & DI H2O Notes: Pipettes & Syringes
(1to 10) L	1 L 3 L 7 L 10 L	0.0072 L 0.027 L 0.067 L 0.097 L	TIF 9010A & DI H2O Notes: Pipettes & Syringes
Force ⁷ – Cable Tension 1/16” 3/32” 1/8” 3/16” 7/32” 1/4” 9/32” 5/16” 3/8”	(5 to 100) lbf (5 to 185) lbf (5 to 352) lbf (10 to 740) lbf (10 to 1000) lbf (20 to 1280) lbf (20 to 1560) lbf (20 to 1800) lbf (20 to 2400) lbf	0.5 % 0.5 % 0.5 % 0.5 % 0.5 % 0.5 % 0.5 % 0.5 % 0.5 %	King Nutronics 3615 Notes: cable tensionmeters, force gages

Parameter/Equipment	Range	CMC ^{2, 6, 7} (±)	Comments
Velocity ³ – Measure	(0 to 300) fpm (0 to 1250) fpm (0 to 2500) fpm (0 to 5000) fpm (0 to 10 000) fpm	2.0 % + 5.1 fpm 2.1 % + 4.7 fpm 2.0 % + 6.0 fpm 2.2 % + 2.0 fpm 2.5 % fpm	Alnor 6006AP
Rotational Speed, RPM ³ – Measure			
Optical	(10 to 100000) rpm	0.001 % + 0.13 rpm	HP 3325A
Mechanical	(0 to 5000) rpm	0.025 %	Quantum Dynamics N-11-ECS/3A

VII. Thermodynamics

Parameter/Equipment	Range	CMC ^{2, 6} (±)	Comments
Relative Humidity ³ – Measuring Equipment	11.31 % RH 33.07 % RH 75.47 % RH 97.6 % RH	0.42 % RH 0.39 % RH 0.44 % RH 0.56 % RH	LiCl MgCl ₂ NaCl K ₂ SO ₄
Relative Humidity ³ – Measure	(10 to 40) % RH (40 to 97) % RH	0.75 % RH 0.99 % RH	Vaisala MI70 w/ HMP-77B
Temperature ³ – Measure			
PRT Probe	(0 to 1000) °C (-196 to 0) °C 0 °C (0 to 232) °C (232 to 420) °C (420 to 660) °C	0.089 °C 0.031 °C 0.01 °C 0.037 °C 0.048 °C 0.34 °C	Fluke 1521, Fluke 5624 PRT, Fluke 5626 PRT
	0 °C (0 to 260) °C	0.025 °C 0.15 °C	Fluke 5627 PRT
Infrared Measure	(0 to 50) °C (50 to 200) °C (200 to 400) °C	1.1 °C 0.6 % + 0.8 °C 0.45 % + 1.1 °C	AR350

Parameter/Equipment	Range	CMC ^{2, 6} (±)	Comments
Temperature ³ – Generate			
Ice point	0 °C	20 mK	DI H ₂ O & shaved ice
Low Temp Bath	-78.5 °C -196 °C	0.5 °C 20 mK	Frozen CO ₂ & ethanol LN2
Drywell	(-25 to 70) °C (70 to 140) °C (50 to 250) °C (250 to 450) °C (450 to 650) °C	0.26 °C - 0.063 % 0.11 % + 0.14 °C 0.38 °C 0.73 °C 0.92 °C	Fluke 9103
Infrared Source	(25 to 500) °C	0.45 % + 0.57 °C	Fluke 9141 NOTE: add 1 °C for wells larger than 6.5 mm Reed BX-500

VIII. Time & Frequency

Parameter/Equipment	Range	CMC ² (±)	Comments
Frequency – Measuring Equipment	10 MHz reference (1 to 10) Hz (10 to 100) Hz (0.1 to 1) kHz (1 to 10) kHz (10 to 100) kHz (0.1 to 1) MHz (1 to 10) MHz (10 to 21) MHz (10 to 100) MHz (0.1 to 1) GHz (1 to 10) GHz (10 to 26.5) GHz	5.5 × 10 ⁻¹¹ Hz/Hz 8.1 × 10 ⁻⁶ Hz/Hz 3.1 × 10 ⁻⁷ Hz/Hz 4.4 × 10 ⁻⁸ Hz/Hz 1.1 × 10 ⁻⁸ Hz/Hz 1.1 × 10 ⁻⁸ Hz/Hz 1.1 × 10 ⁻⁹ Hz/Hz 1.4 × 10 ⁻¹⁰ Hz/Hz 8.0 × 10 ⁻¹¹ Hz/Hz 2.1 × 10 ⁻⁸ Hz/Hz 1.2 × 10 ⁻⁹ Hz/Hz 1.2 × 10 ⁻¹⁰ Hz/Hz 8.0 × 10 ⁻¹¹ Hz/Hz	Agilent Z3816A Agilent 3325B Agilent 83630A
Frequency – Measure	1 Hz to 225 MHz (10 to 100) MHz (0.1 to 1) GHz (1 to 10) GHz (10 to 20) GHz	5.6 × 10 ⁻¹¹ Hz/Hz 1.9 × 10 ⁻⁸ Hz/Hz 1.8 × 10 ⁻⁹ Hz/Hz 1.6 × 10 ⁻¹⁰ Hz/Hz 1.2 × 10 ⁻¹⁰ Hz/Hz	Agilent 53132A HP 5350B

¹ This laboratory offers commercial calibration service and field calibration services.

² Calibration and Measurement Capability Uncertainty (CMC) is the smallest uncertainty of measurement that a laboratory can achieve within its scope of accreditation when performing more or less routine calibrations of nearly ideal measurement standards or nearly ideal measuring equipment. CMCs represent expanded uncertainties expressed at approximately the 95 % level of confidence, usually using a coverage factor of $k = 2$. The actual measurement uncertainty of a specific calibration performed by the laboratory may be greater than the CMC due to the behavior of the customer's device and to influences from the circumstances of the specific calibration.

³ Field calibration service is available for this calibration and this laboratory meets A2LA *R104 – General Requirements: Accreditation of Field Testing and Field Calibration Laboratories* for these calibrations. Please note the actual measurement uncertainties achievable on a customer's site can normally be expected to be larger than the CMC found on the A2LA Scope. Allowance must be made for aspects such as the environment at the place of calibration and for other possible adverse effects such as those caused by transportation of the calibration equipment. The usual allowance for the actual uncertainty introduced by the item being calibrated, (e.g. resolution) must also be considered and this, on its own, could result in the actual measurement uncertainty achievable on a customer's site being larger than the CMC.

⁴ The stated measured values are determined using the indicated instrument (see Comments). This capability is suitable for the calibration of the devices intended to measure or generate the measured value in the ranges indicated. CMCs are expressed as either a specific value that covers the full range or as a fraction/percentage of the reading plus a fixed floor specification.

⁵ In the statement of CMC, L is the numerical value of the nominal length of the device measured in inches, and D is the diameter of the device in inches.

⁶ In the statement of CMC, percentages are percentages of reading, unless otherwise indicated.

⁷ The contributions from the "best existing device" are not included in the CMC claim.



Accredited Laboratory

A2LA has accredited

ACR TECHNICAL SERVICES, INC.

Newport News, VA

for technical competence in the field of

Calibration

This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005 *General requirements for the competence of testing and calibration laboratories*. This laboratory also meets the requirements of ANSI/NCSL Z540-1-1994, ANSI/NCSL Z540.3-2006, and any additional program requirements in the field of calibration. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communiqué dated 8 January 2009).



Presented this 6th day of April, 2015.

A handwritten signature in black ink, reading 'Peter Meyer'.

President & CEO

For the Accreditation Council

Certificate Number 2171.01

Valid to March 31, 2017

Revised February 21, 2017

For the calibrations to which this accreditation applies, please refer to the laboratory's Calibration Scope of Accreditation.